CHAPTER EXERCISE: Answers for this chapter start on page 272.

A calculator should NOT be used on the following questions.

1

If $a^{-\frac{1}{2}} = 3$, what is the value of a ?

A) -9

B) $\frac{1}{9}$

C) $\frac{1}{3}$

D) 9

2

If $\frac{2^x}{2^y} = 2^3$, then x must equal

A) $y+3$

B) $y-3$

C) $3-y$

D) $3y$

3

If $y^5 = 10$, what is the value of y^{20} ?

A) 40

B) 400

C) 1,000

D) 10,000

4

The expression $\sqrt[4]{x^2y^4}$, where $x > 0$ and $y > 0$, is equivalent to which of the following?

A) $\sqrt{xy}$

B) $y\sqrt{x}$

C) $\frac{1}{x^2}$

D) x^2y

5

If $\frac{\sqrt{x^3}}{\sqrt[4]{x}} = x^c$ for all positive values of x, what is the value of c ?

6

If $3^x = 10$, what is the value of 3^{x-3} ?

A) $\frac{10}{3}$

B) $\frac{10}{9}$

C) $\frac{10}{27}$

D) $\frac{27}{10}$

7

If a and b are positive even integers, which of the following is greatest?

A) $(-2a)^b$

B) $(-2a)^{2b}$

C) $(2a)^b$

D) $2a^{2b}$

8

If $x^2 = y^3$, for what value of z does $x^{3z} = y^9$?

A) -1

B) 0

C) 1

D) 2

9

If $\sqrt{x\sqrt{x}} = x^a$, then what is the value of a ?

A) $\frac{1}{2}$

B) $\frac{3}{4}$

C) 1

D) $\frac{4}{3}$

10

If $x^{ac} \cdot x^{bc} = x^{30}$, $x > 1$, and $a + b = 5$, what is the value of c ?

A) 3

B) 5

C) 6

D) 10

11

If $4^{2n+3} = 8^{n+5}$, what is the value of n ?

A) 6

B) 7

C) 8

D) 9

12

Which of the following is equivalent to $(-2)^{\frac{5}{3}}$?

A) $-2 \cdot \sqrt[3]{4}$

B) $2 \cdot \sqrt[3]{4}$

C) $-4 \cdot \sqrt[3]{2}$

D) $4 \cdot \sqrt[3]{2}$

13

If $2^{x+3} - 2^x = k(2^x)$, what is the value of k ?

A) 3

B) 5

C) 7

D) 8

A calculator is allowed on the following questions.

14

If $(5^3)^{4k} = (5^{\frac{1}{3}})^{24}$, what is the value of k ?

A) -6

B) $\frac{2}{3}$

C) $\frac{3}{4}$

D) 2

15

Which of the following is equivalent to $x^{\frac{2a}{b}}$ for all positive values of x, where a and b are positive integers?

A) $\sqrt[b]{ax^2}$

B) $\sqrt[b]{x^{2a}}$

C) $\sqrt[b]{x^{a+2}}$

D) $\sqrt[2a]{x^b}$

16

If $x^2y^3 = 10$ and $x^3y^2 = 8$, what is the value of x^5y^5 ?

A) 18

B) 20

C) 40

D) 80

CHAPTER EXERCISE:

1. $\boxed{B}$

$$a^{-\frac{1}{2}} = 3$$
$$\frac{1}{a^{\frac{1}{2}}} = 3$$
$$1 = 3\sqrt{a}$$
$$\frac{1}{3} = \sqrt{a}$$
$$\frac{1}{9} = a$$

2. $\boxed{A}$

$$\frac{2^x}{2^y} = 2^3$$
$$2^{x-y} = 2^3$$
$$x - y = 3$$
$$x = y + 3$$

3. $\boxed{D}$ Raise each side to the 4th power:

$$y^5 = 10$$
$$(y^5)^4 = 10^4$$
$$y^{20} = 10{,}000$$

4. $\boxed{B}$

$$\sqrt[4]{x^2y^4} = (x^2y^4)^{\frac{1}{4}} = x^{2\cdot\frac{1}{4}}y^{4\cdot\frac{1}{4}} = x^{\frac{1}{2}}y^1 = y\sqrt{x}$$

5. $\boxed{\frac{5}{4}}$ $\frac{\sqrt{x^3}}{\sqrt[4]{x}} = \frac{x^{\frac{3}{2}}}{x^{\frac{1}{4}}} = x^{\frac{3}{2}-\frac{1}{4}} = x^{\frac{6}{4}-\frac{1}{4}} = x^{\frac{5}{4}}$.

Therefore, $c = \frac{5}{4}$.

6. $\boxed{C}$

$$3^{x-3} = \frac{3^x}{3^3} = \frac{10}{3^3} = \frac{10}{27}$$

7. $\boxed{B}$ To avoid any trickiness, it's best to plug in numbers. Let $a = 2$ and $b = 2$. Going through each choice,

A) $(-4)^2 = 16$

B) $(-4)^4 = 256$

C) $(2 \cdot 2)^2 = 16$

D) $2 \cdot 2^4 = 2 \cdot 16 = 32$

(B) is the largest.

8. $\boxed{D}$ Cube both sides of the first equation,

$$(x^2)^3 = (y^3)^3$$
$$x^6 = y^9$$

Now y^9 can be replaced by x^6,

$$x^{3z} = y^9$$
$$x^{3z} = x^6$$
$$3z = 6$$
$$z = 2$$

9. $\boxed{B}$

$$\sqrt{x\sqrt{x}} = \sqrt{x \cdot x^{\frac{1}{2}}} = \sqrt{x^{\frac{3}{2}}} = (x^{\frac{3}{2}})^{\frac{1}{2}} = x^{\frac{3}{4}}$$

Therefore, $a = \frac{3}{4}$

10. $\boxed{C}$

$$x^{ac} \cdot x^{bc} = x^{30}$$
$$x^{ac+bc} = x^{30}$$
$$ac + bc = 30$$
$$(a + b)c = 30$$
$$5c = 30$$
$$c = 6$$

11. $\boxed{D}$

$$\begin{aligned} 2^{2(2n+3)} &= 2^{3(n+5)} \\ 2(2n+3) &= 3(n+5) \\ 4n+6 &= 3n+15 \\ n &= 9 \end{aligned}$$

12. $\boxed{A}$ $(-2)^{\frac{5}{3}} = \sqrt[3]{(-2)^5} =$
$\sqrt[3]{\boxed{-2\cdot -2\cdot -2}\cdot -2\cdot -2} = -2\cdot\sqrt[3]{-2\cdot -2} =$
$-2\cdot\sqrt[3]{4}$

13. $\boxed{C}$

$$\begin{aligned} 2^{x+3} - 2^x &= k(2^x) \\ (2^x)(2^3) - 2^x &= k(2^x) \\ 2^x(2^3-1) &= k(2^x) \\ 2^x(7) &= k(2^x) \\ 7 &= k \end{aligned}$$

14. $\boxed{B}$ Multiply the exponents.

$$\begin{aligned} (5^3)^{4k} &= (5^{\frac{1}{3}})^{24} \\ 5^{12k} &= 5^8 \end{aligned}$$

Since the bases are the same, we can equate the exponents: $12k = 8$ and so $k = \frac{8}{12} = \frac{2}{3}$.

15. $\boxed{B}$ The $2a$ means raised to the $2a$ power and the b on the bottom means the bth root.

16. $\boxed{D}$ Multiply both equations together. The left hand side gives x^5y^5. The right hand side gives 80.